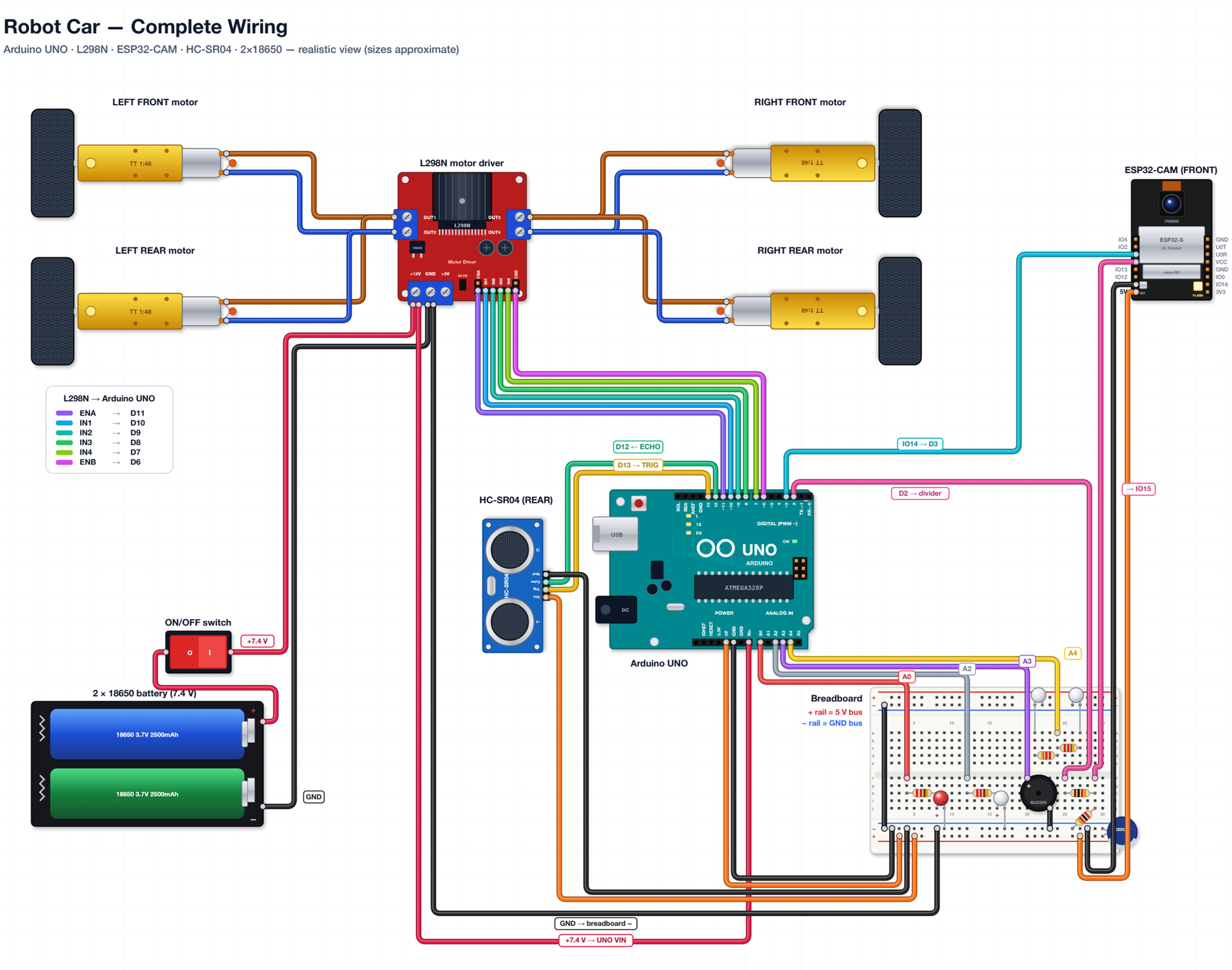
ROBOT CAR PROJECT · WIRING POSTER

Complete wiring: realistic view

L298N ENA–ENB → D11–D6 · TRIG D13 · ECHO D12 · LEDs A0/A2 · buzzer A3 · headlights A4 · IO14→D3 · D2→1k→IO15 (2k to GND)



⚠ **Power:** battery 7.4 V → switch → L298N +12V, and a 2nd wire from that terminal → UNO VIN. The UNO 5V pin feeds the 5 V rail. 2200 µF at the ESP32-CAM.

Remove the ENA/ENB jumpers on the L298N. Keep the 5V-EN jumper.

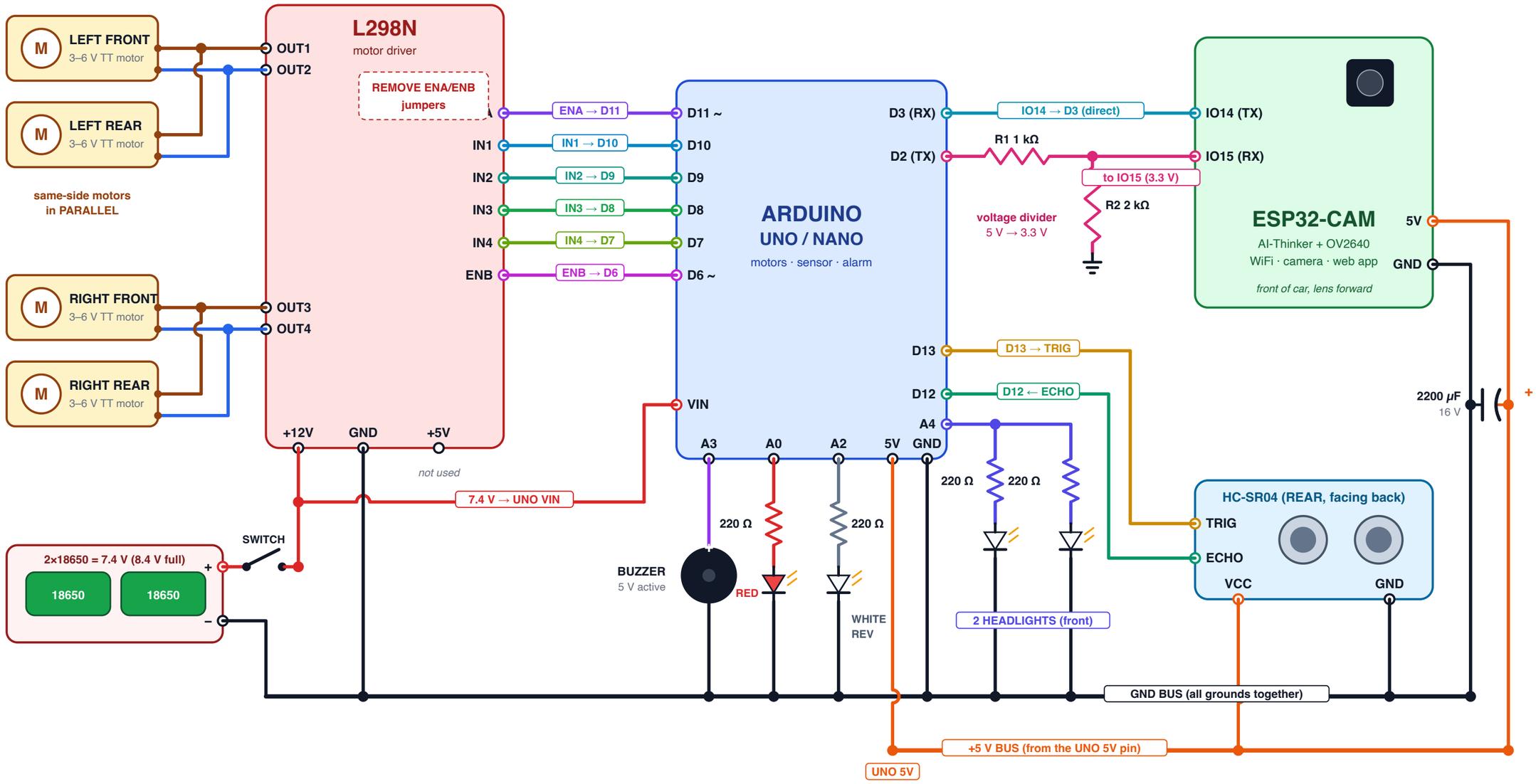
All GNDs common. 7.4 V only to L298N +12V and UNO VIN, never to a 5V pin.

ROBOT CAR PROJECT · WIRING POSTER

Complete wiring: schematic view

ROBOT CAR – SCHEMATIC WIRING DIAGRAM

4WD chassis · L298N · Arduino UNO/Nano · ESP32-CAM · HC-SR04 (rear) · buzzer + rear LEDs · 2 headlights · 2×18650 (no buck converter: the UNO makes the 5 V)



POWER — no buck converter needed

- Battery 7.4 V → switch → L298N +12V. A 2nd wire in the same +12V terminal → UNO VIN.
- The UNO regulator makes the 5 V: UNO 5V pin → 5 V bus (ESP32-CAM, HC-SR04, buzzer, LEDs).
- 2200 µF 16 V capacitor right at the ESP32-CAM 5V/GND pins (+ leg to 5V).
- L298N: keep the 5V-EN jumper ON, remove ENA/ENB jumpers, leave its +5V pin unused.
- Do not add a servo to this 5 V bus (the UNO regulator is too small for it).

WIRE COLOURS

Battery +7.4 V	+5 V	GND	Motor wires	UART ESP→UNO
UART UNO→ESP	TRIG	ECHO	Buzzer	Headlights

- Wires crossing with a small hop are NOT connected. A black dot = connected.
- **ALL grounds must be common. Never feed 7.4 V into the ESP32-CAM or UNO 5V pin.**